

Sectoral differences in the heat sensitivity of electricity consumption in Hong Kong

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Abstract

This assignment uses a new analysis of official historical observations to examine sectoral differences in Hong Kong’s heat–electricity relationship. Hong Kong Observatory daily temperatures and Census and Statistics Department monthly electricity data cover January 2010–December 2025. Regressions for 192 months account for calendar-month and year effects, while an exploratory network of 28 weather stations describes spatial temperature patterns. Ten additional cooling degree-days above 24°C are associated with 1.84% higher total consumption and 3.50% higher residential consumption. Sector comparisons support a stronger residential association. The models provide evidence of differentiated heat sensitivity, grounding a complex-systems discussion of heterogeneous urban activities. Feedback, adaptation and interactions among users remain mechanisms for further investigation.

Introduction

Urban heat links thermal comfort to citywide electricity consumption. Air conditioning helps households and businesses cope with hot conditions, while its use adds to aggregate energy demand (Davis and Gertler, 2015). In Hong Kong, residential, commercial and industrial

activities may therefore respond differently to the same thermal conditions. This heterogeneity provides an entry point for viewing the city as a complex system: shared environmental pressure interacts with diverse activities to generate collective patterns of resource use. Figure 1 sketches the feedback pathways considered in this paper.

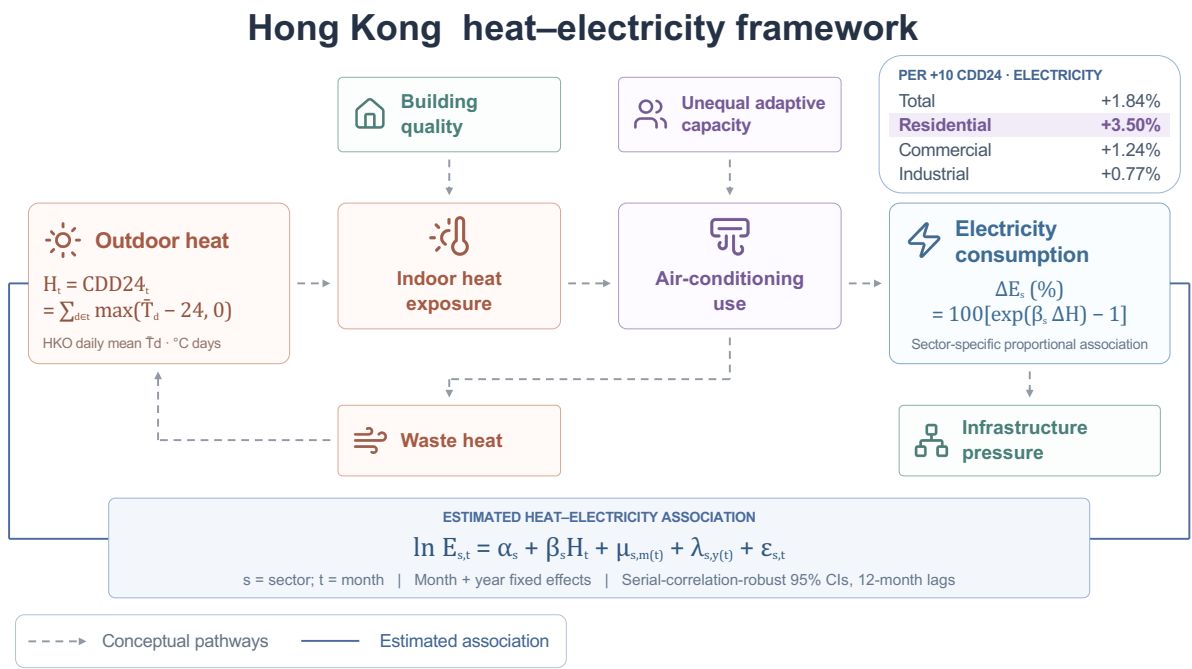


Figure 1. Conceptual heat and electricity interactions. The solid connection denotes the estimated heat–electricity association. Dashed arrows represent proposed pathways involving buildings, cooling, waste heat, adaptive capacity and infrastructure pressure; these mechanisms were not tested. Calculations for this assignment use Hong Kong Observatory and Census and Statistics Department observations for January 2010–December 2025, downloaded on 23 September 2026 (Hong Kong time).

This assignment asks whether heat is associated with electricity consumption beyond recurring seasonality, and whether that association differs across residential, commercial and industrial activities. Official historical observations were assembled, processed and modelled for this study. Calendar-month and year controls distinguish the remaining heat–electricity association from regular seasonal cycles and annual level differences. Comparing sector coefficients then tests whether a common heat indicator is associated with heterogeneous proportional changes in consumption. These questions

make the empirical contribution explicit and connect the modelling to the complex-systems concern with diversity within aggregate urban patterns.

Spatial heterogeneity adds a second layer of complexity. Urban form influences local ventilation in Hong Kong (Ng, 2009), and stations can share day-to-day fluctuations while maintaining different baseline warmth. To situate citywide electricity associations within this uneven thermal landscape, Figure 2 summarises the temperature correlation structure across stations and highlights warm-season mean differences.



Figure 2. Temperature correlation network. The 28 stations have $\geq 90\%$ complete daily-mean coverage. Colours show May–September mean temperatures. Links comprise 27 maximum-spanning-tree edges and 13 additional correlations ≥ 0.98 after seasonal adjustment. Links indicate statistical similarity. Sources: Hong Kong Observatory, Lands Department and Home Affairs Department. Temperature observations cover 1 January 2010–31 December 2025. Station calculations were produced for this assignment. Weather and geographic data snapshots were retrieved on 23 September 2026 (Hong Kong time).

Local evidence

The empirical analysis was conducted for this assignment using official secondary data downloaded on 23 September 2026 (Hong Kong time). Monthly local electricity consumption, in terajoules, came from the Census and Statistics Department’s Table 915-91201; daily mean, maximum and minimum temperatures came from the Hong Kong Observatory. The analysis covers January 2010–December 2025, yielding 192 monthly observations for each electricity series. Total, residential, commercial and industrial consumption were analysed separately, using Observatory headquarters temperature

as a common exposure proxy. Data cleaning, monthly heat indicators, regressions and the station network were produced within this project. Figure 3 describes CDD24 and electricity seasonality and their month-demeaned anomalies.

CDD24 sums $\max(\text{daily mean temperature} - 24^\circ\text{C}, 0)$ within each month; very hot days and hot nights count days with maxima $\geq 33^\circ\text{C}$ and minima $\geq 28^\circ\text{C}$, respectively. The main indicators use observations flagged complete by the Observatory, without interpolation or zero-filling. Separate regressions of log consumption include one heat indicator at a time, calendar-month ef-

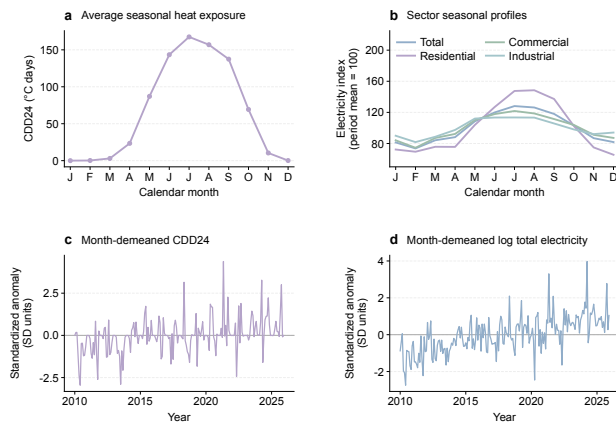


Figure 3. Seasonality and descriptive anomalies.

Panels a–b show monthly climatologies; electricity is indexed to each series mean of 100. CDD24 denotes cooling degree-days above 24°C. Panels c–d show standardised month-demeaned anomalies, not month-and-year regression residuals. Calculations for this assignment use Hong Kong Observatory and Census and Statistics Department observations for January 2010–December 2025, downloaded on 23 September 2026 (Hong Kong time).

fects and year effects. Month effects account for recurring seasonality, while year effects account for annual level differences. Newey–West standard errors with 12 calendar-month lags allow for heteroskedasticity and serial correlation. The models assess adjusted associations and sector differences; humidity, economic activity and billing timing may still confound those associations.

After adjustment, CDD24 remains positively associated with consumption in all four series, with different estimated slopes across sectors. Figure 4 plots relationships after removing month and year effects from both variables; Figure 5 reports proportional associations and their uncertainty. Industrial estimates for very hot days and hot nights have intervals containing zero, so the evidence varies across heat indicators.

Ten additional CDD24 degree-days are associated with 1.84% higher total consumption (95% confidence interval, 1.68–1.99%) and 3.50% higher residential consumption (3.09–3.92%), compared with 1.24% for commercial and 0.77% for industrial consumption. Models of residential-to-commercial and residential-to-industrial consumption ratios also support larger residential coefficients. These estimates describe proportional differences within each series, rather than sectors' absolute contributions to demand. Their relevance to complex-systems interpretation lies in measurable heterogeneity: the citywide association contains distinct sectoral relationships with the same heat indicator. Occupancy, building conditions and electricity end uses offer possible explanations, but the models do not identify these mechanisms or interactions among consumers.

The station analysis uses daily observations from 2010–2025 and provides spatial context for the territory-wide electricity models. Of 32 stations examined, 28 met the

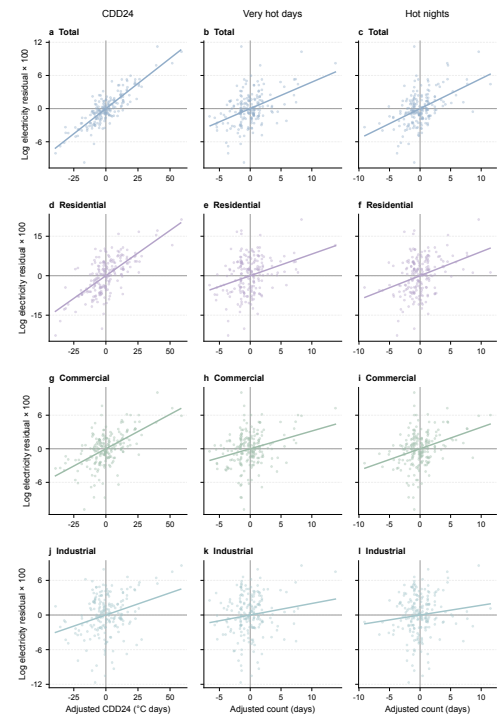


Figure 4. Adjusted heat and electricity associations.

Both axes remove month and year effects across 192 months. Log-consumption residuals multiplied by 100 are not exact percentage changes. Very hot days have maximum temperature $\geq 33^\circ\text{C}$; hot nights have minimum temperature $\geq 28^\circ\text{C}$. Calculations for this assignment use Hong Kong Observatory and Census and Statistics Department observations for January 2010–December 2025, downloaded on 23 September 2026 (Hong Kong time).

$\geq 90\%$ complete daily-mean coverage criterion. After removing each station's calendar-day climatology, pairwise correlations range from 0.860 to 0.993, while warm-season mean temperatures differ (Figure 2). Shared fluctuations therefore coexist with local thermal differences. The network describes statistical similarity; its connectivity is imposed by the maximum spanning tree and does not establish physical heat transfer. Only two stations meet the $\geq 90\%$ complete daily-minimum criterion. Supplementary Figure S1 distinguishes this strict sample from the exploratory 29-station sample that includes incomplete numeric minima.

The positive associations persist when the cooling-degree-day threshold changes from 24°C to 26°C , when 2020–2022 is excluded, and when March–April 2023 is excluded because the official source identifies a billing-timing anomaly. These checks support the stability of the observed pattern under the tested specifications. Together, the analyses establish an adjusted heat–electricity association, sectoral differences and spatial variation in temperature. The following discussion uses these findings to examine complex-systems explanations, distinguishing measured patterns from mechanisms that require additional evidence.

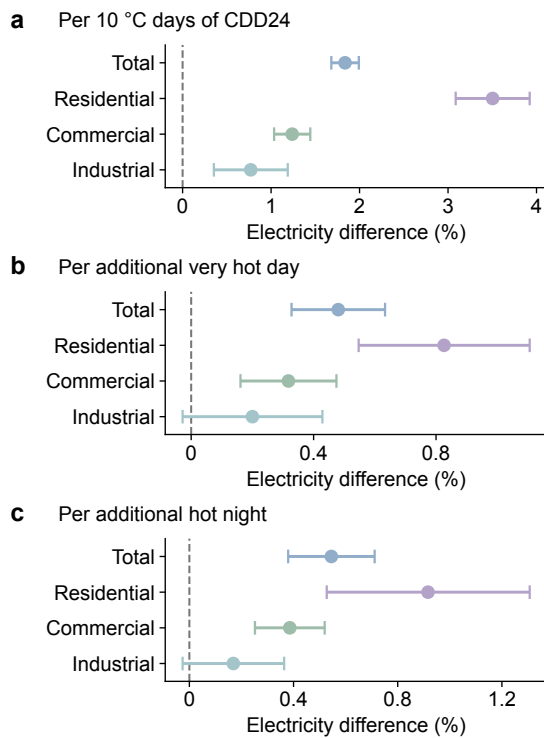


Figure 5. Sectoral associations and uncertainty. Bars show 95% confidence intervals using 12 calendar-month lags. Increments are ten CDD24 degree-days (a), one very hot day (b) and one hot night (c). Calculations for this assignment use Hong Kong Observatory and Census and Statistics Department observations for January 2010–December 2025, downloaded on 23 September 2026 (Hong Kong time).

Complex systems interpretation

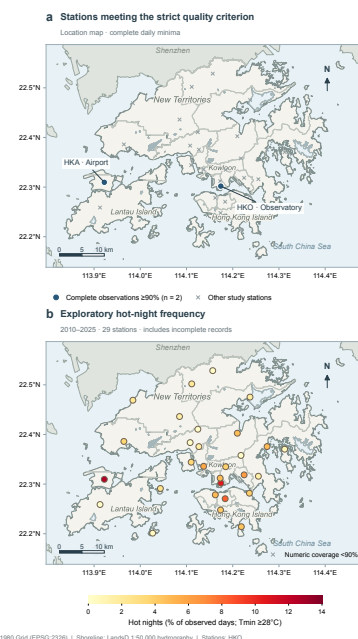
Emergence concerns how city-scale patterns arise from decisions and interactions at smaller scales (Batty, 2012). The sector models contribute to this question by showing that aggregate heat sensitivity contains different residential, commercial and industrial relationships. A single total-consumption coefficient would conceal this variation. Local cooling decisions provide one plausible route from thermal conditions to collective demand, with equipment, buildings and routines shaping different responses. However, common weather can also generate correlated consumption without interactions among users. The analysis therefore identifies heterogeneous aggregate patterns, while testing emergence through interactions would require linked household or building observations. Its contribution is to make the pattern requiring explanation measurable.

Interdependence connects these decisions to the urban environment. Building materials, surface moisture and urban geometry influence heat storage and exchange, linking local temperatures to the arrangement of surrounding surfaces (Oke, 1982). In Hong Kong, air ventilation assessment similarly recognises that development layouts shape airflow through the urban fabric (Ng, 2009). Applied to electricity use, these mechanisms suggest that a household's cooling requirements may depend partly on neighbouring buildings

and planning decisions beyond its control. The relevant interactions therefore cross both property boundaries and sectors: urban development helps shape thermal conditions, while occupants respond within those conditions. This perspective offers a plausible explanation for the persistence of heat-related demand, because changing individual behaviour alone leaves the surrounding physical constraints in place.

Feedback arises when responses to heat alter the conditions that prompted them. Temperature-controlled cooling can counteract indoor warming, forming a balancing feedback, while rejected heat may raise outdoor temperatures and encourage further cooling, forming a reinforcing loop. Simulations for Phoenix demonstrate the latter possibility, particularly at night (Salamanca et al., 2014). Figure 1 places the estimated Hong Kong heat–electricity association within this wider conceptual framework. The regression measures one association relevant to the proposed loop; it does not measure air-conditioning use, waste heat or the resulting temperature change. Establishing feedback locally would require observations or validated simulations linking these processes over time. The present models thus motivate a specific systems hypothesis while leaving its operation and magnitude unresolved.

Adaptation can also change the relationship between heat and electricity over time. Households may adjust operating hours, replace equipment or improve



Supplementary Figure S1. Hot-night exposure and data coverage. Panel a locates two stations with $\geq 90\%$ complete-minimum coverage. Panel b includes 29 stations with $\geq 90\%$ numeric coverage, retaining incomplete records. Colours indicate observed hot-night frequency, not population vulnerability. Sources: Hong Kong Observatory, Lands Department and Home Affairs Department. Temperature observations cover 1 January 2010–31 December 2025. Station calculations were produced for this assignment. Weather and geographic data snapshots were retrieved on 23 September 2026 (Hong Kong time).

buildings, so future demand depends on how responses evolve. Research in Mexico distinguishes temperature-driven changes in electricity use from longer-term changes in air-conditioning ownership (Davis and Gertler, 2015). Efficiency improvements can introduce a further behavioural response: cheaper cooling may encourage longer operation, partly offsetting expected savings. In a Mexican appliance replacement programme, air-conditioner replacement increased electricity consumption despite the introduction of more efficient equipment (Davis et al., 2014). This result illustrates why technical efficiency and total consumption need not move together. For Hong Kong, the implication is to evaluate actual electricity use alongside comfort improvements after interventions. The 2010–2025 coefficients summarise historical associations; they cannot show whether adaptation strengthened or weakened heat sensitivity over time.

These adjustments are also path dependent, because earlier investments shape which responses remain practical. Research on technological lock-in explains how infrastructure and supporting institutions can reinforce established energy systems (Unruh, 2000). Applied to Hong Kong, this suggests that existing building layouts and installed equipment may favour continued mechanical cooling. Replacing a cooling unit can be easier than changing a building's orientation or the surrounding street layout. Successive equipment upgrades may therefore preserve reliance on electricity even as individual units become more efficient. This mechanism could help explain persistent heat-sensitive demand, although the sectoral regressions do not test it. Renovation and redevelopment provide opportunities to reconsider these constraints through coordinated changes to buildings and their surroundings. Adaptation consequently involves choices at different timescales: adjustments to daily operation address immediate discomfort, while decisions about buildings and urban form shape the conditions under which future demand develops.

Implications and limitations

These mechanisms suggest combining efficient equipment with measures that reduce heat exposure. Hong Kong simulations found that tree planting could cool pedestrian areas, whereas roof greening offered limited cooling near the ground in tall urban settings (Ng et al., 2012). Modelling in Seoul indicates that green roofs can reduce building energy use, illustrating that benefits depend on the outcome, urban setting and spatial scale assessed (Adilkanova et al., 2024). For Hong Kong, measures should therefore be selected for particular buildings and streets, with ventilation, shading and electricity use evaluated together. The stronger residential association makes housing a relevant focus, but does not identify which neighbourhoods or interventions would yield the largest benefits. Evaluation should track indoor comfort alongside electricity consumption, so that reduced consumption is not automatically treated as successful adaptation.

References

- I. Adilkanova, M. Santamouris, and G. Y. Yun. Green roofs save energy in cities and fight regional climate change. *Nature Cities*, 1:238–249, 2024. 10.1038/s44284-024-00035-7.
- M. Batty. Building a science of cities. *Cities*, 29(Supplement 1):S9–S16, 2012. 10.1016/j.cities.2011.11.008.
- L. W. Davis and P. J. Gertler. Contribution of air conditioning adoption to future energy use under global warming. *Proceedings of the National Academy of Sciences*, 112(19):5962–5967, 2015. 10.1073/pnas.1423558112.
- L. W. Davis, A. Fuchs, and P. Gertler. Cash for coolers: Evaluating a large-scale appliance replacement program in Mexico. *American Economic Journal: Economic Policy*, 6(4):207–238, 2014. 10.1257/pol.6.4.207.
- W. B. Goggins, E. Y. Y. Chan, E. Ng, C. Ren, and L. Chen. Effect modification of the association between short-term meteorological factors and mortality by urban heat islands in Hong Kong. *PLOS ONE*, 7(6):e38551, 2012. 10.1371/journal.pone.0038551.
- E. Ng. Policies and technical guidelines for urban planning of high-density cities – air ventilation assessment (AVA) of Hong Kong. *Building and Environment*, 44(7):1478–1488, 2009. 10.1016/j.buildenv.2008.06.013.
- E. Ng, L. Chen, Y. Wang, and C. Yuan. A study on the cooling effects of greening in a high-density city: An experience from Hong Kong. *Building and Environment*, 47:256–271, 2012. 10.1016/j.buildenv.2011.07.014.
- T. R. Oke. The energetic basis of the urban heat island. *Quarterly Journal of the Royal Meteorological Society*, 108(455):1–24, 1982. 10.1002/qj.49710845502.
- F. Salamanca, M. Georgescu, A. Mahalov, M. Moustauoui, and M. Wang. Anthropogenic heating of the urban environment due to air conditioning. *Journal of Geophysical Research: Atmospheres*, 119(10):5949–5965, 2014. 10.1002/2013JD021225.
- G. C. Unruh. Understanding carbon lock-in. *Energy Policy*, 28(12):817–830, 2000. 10.1016/S0301-4215(00)00070-7.

Official data sources

Census and Statistics Department, Hong Kong SAR. Table 915-91201: monthly local electricity consumption by user category, in terajoules. Observations analysed: January 2010–December 2025. Data retrieved 23 September 2026 (Hong Kong time). https://www.censtatd.gov.hk/en/web_table.html?id=915-91201

Hong Kong Observatory. Daily climate data: daily mean, maximum and minimum air temperatures. Observations analysed: 1 January 2010–31 December 2025. Data retrieved 23 September 2026 (Hong Kong time). <https://www.hko.gov.hk/en/cis/downloadpage.htm>

Hong Kong Observatory. Station information: coordinates, elevations and operating dates used to document the study stations. Source snapshot retrieved 23 September 2026 (Hong Kong time). Station metadata.

Home Affairs Department and Lands Department, Hong Kong SAR. District boundaries and HYDRPOLY hydrography, accessed through the Common Spatial Data Infrastructure portal. Geographic snapshots used as map

backgrounds; retrieved 23 September 2026 (Hong Kong time). District boundaries; Hydrography.

Code availability

The code used in this study are publicly available in the GitHub repository `LSGI2802_Assignment1`. The repository is available at https://github.com/Claire0712/LSGI2802_Assignment1.

AI Declaration

This report has used AI-assisted tools for language refinement, structural editing, and formatting support. All arguments, interpretations, data selection, and final academic judgments are the sole responsibility of the author.